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Prompt engineering and generative AI in the academic workflow

AI in research work · parts 2 and 3

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Prompt engineering

A?

“the process of constructing queries or inputs for AI language models so as to elicit the most precise, coherent, and pertinent responses”

The definition this course uses — wording varies across sources.

A prompt is a question with the context baked in

PROMPT

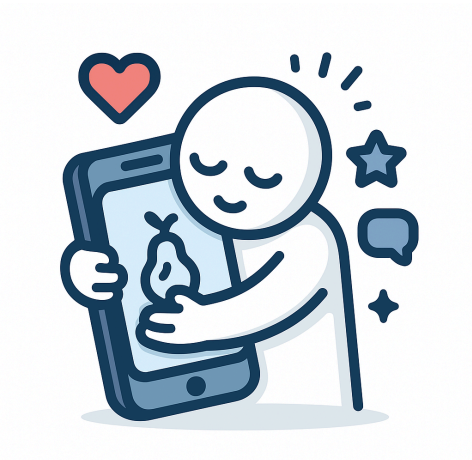
What is love?



sociology



chemistry



business



music

social interactions · dopamine and norepinephrine · attachment to a brand · “oh baby, don’t hurt me”

> The model did not change its mind about the words — it answered the question it could infer. Give it the context, and you get the answer you meant.

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Be clear, give context, say what you want

Principle	The prompt you typed	The prompt that works
Clear and concise	“Tell me about the climate”	Summarize the key impacts of climate change observed globally between 2015 and 2025, highlighting effects on agriculture.
Contextual	“What is PCE?”	What is PCE in the context of photovoltaics?
Instructive	“Analyze the following text: (BLOCK OF TEXT)”	Provide a short summary of the following text (50-70 words) explaining the reported key concepts. Provide a list with the main results, itemize the different methods, and explain the novelty.

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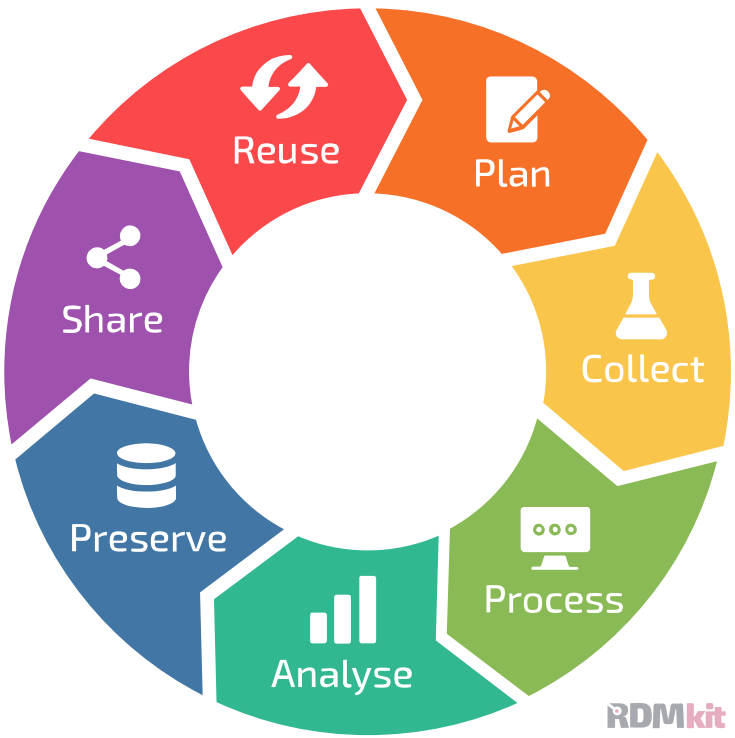
Split the task, give it a role, set the tone

Principle	The prompt you typed	The prompt that works
Iterative	“Write about renewable energies”	List four types … → from that list, describe the advantages and drawbacks of solar → expand on the environmental impacts of large-scale solar.
Roleplay	“Explain to me the basics of antibiotics.”	You are an experienced microbiology teacher who makes explanations accessible and fun for students. Explain how misuse of antibiotics contributes to resistance.
Tone and format	“Tell me about black holes.”	Explain black holes in a friendly tone suitable for toddlers, with simple analogies, in bullet points, and briefly explain how the analogy works.

Generative AI in the academic workflow

A?

Every phase of the research process has an AI use



Phase	What people actually use it for
Plan	grant writing, research planning, study design
Collect	collection interfaces, code and automation help
Process	preprocessing ideas, automation, data minimisation, annotation
Analyse	annotations, methodological choices, analysis code
Preserve	metadata preparation, summarising, organising
Share · Reuse	discovery and FAIR-ness, open datasets, research gaps
Write	editing, tables and figures, peer review, public summaries

When the answer is wrong, fix the prompt — not the model

PROMPT

Make a prompt for a Google Scholar search to find articles about azobenzene and spinning.

AI

“Spinning properties of azobenzene in scholarly articles”

When the answer is wrong, fix the prompt — not the model

PROMPT

Make a prompt for a Google Scholar search to find articles about azobenzene and spinning.

AI

“Spinning properties of azobenzene in scholarly articles”

PROMPT

No, the prompt is to be used in the Google Scholar website. Create it so it can efficiently search for articles that contain both "azobenzene" and "spinning" terms, with all their variations.

AI

```
(“Azobenzene” OR “Azo-benzene” OR “Azo benzene”) AND (“Spinning” OR “Spin” OR “Rotation” OR “Rotatory”)
```

Round 1 was a sentence about the topic, not a query.

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It turns prose into documentation

manual text · markdown

PROMPT

Create a markdown readme for an experiment in the equipment with the following description: "Dynamic mechanical analysis (...) this approach can be used to locate the glass transition temperature of the material, as well as to identify transitions corresponding to other molecular motions."

```
# Dynamic Mechanical Analysis Experiment README
## Experiment Name
Dynamic Mechanical Analysis (DMA)
## Introduction
… understanding the viscoelastic behavior of polymers.  (…)
## Objective
1. Determine the complex modulus of the material.  (…)
```

The description already existed — in the manual. This is formatting, and formatting is what it is good at.

It writes the code you would otherwise look up

software with good docs online | code block

PROMPT

In LAMMPS, what pair-wise potential can I use for simulating the interaction between hard-like sphere particles? Give an example of the command.

AI

Use the Weeks–Chandler–Andersen (WCA) potential: Lennard-Jones truncated and shifted at its minimum.

```
pair_style lj/cut 1.122462048309373
pair_coeff * * 1.0 1.0
pair_modify shift yes
```

The cutoff is 2 raised to 1/6 times sigma — right, but check it: it invented that number from a memory of the formula, not from your force field.

One question, two tools: name the software you are in

PROMPT

In Python, how to calculate the 1st and the 3rd quartile of data?
Provide a working example.

AI

Use `numpy`'s `percentile` — it interpolates by default.

```
import numpy as np
data = [1,2,3,4,5,6,7,8,9,10]
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
```

PROMPT

In Excel, how to calculate the 1st and the 3rd quartile of data?
Provide a working example.

AI

Use `QUARTILE.INC` on the range; the second argument is the quartile.

```
=QUARTILE.INC(A2:A11, 1)
=QUARTILE.INC(A2:A11, 3)
```


old slides 31 + 32 · CC-BY 10.5281/zenodo.14032261

It is a brainstorming partner for methods, not an oracle

statistics · named the failure mode

PROMPT

I've been using a moving average technique to fit my dataset, but I'm encountering issues with inaccuracies at the edges of the data. Could you recommend alternative data fitting methods that might provide more reliable results across the entire dataset?

AI

- Polynomial regression · exponential smoothing · LOESS/LOWESS
- Cubic or B-splines · kernel regression · running median
- and: cross-validate, or you will overfit the edges you just fixed

Ask it for a list, then read about the two you have never heard of. It is a table of contents with a bad memory.

Ask for the critique, not the rewrite

keep my voice | teach me to fish

PROMPT

I've written a paragraph that I want to refine and improve in terms of clarity, coherence, and effectiveness, while maintaining my own writing style. (...) Could you provide a detailed critique? Identify weaknesses and explain why they hinder the writing — with suggestions on how to fix them without changing the fundamental style of my writing.

AI

It returns the diagnosis **and** one surgical example: “The initial brown awns become light yellowish, indicating that lignin was somewhat removed, Fig. 5(B).” → “The initial brown awns turned light yellowish, indicating partial removal of lignin (Fig. 5B).”

Written by AI alert: “The initially tawny awns transitioned to a pale yellow hue, signaling a substantive diminution of lignin content.” — that is what happens when you ask for the fish.

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Name the audience, the tone and the format

a specific network | tone | style

PROMPT

I want to share my work on Instagram. Make the language more accessible, write in the style of an influencer, add some emojis, and add relevant hashtags at the end. Use the following abstract: "Active fabrics, responding autonomously to environmental changes (...)."

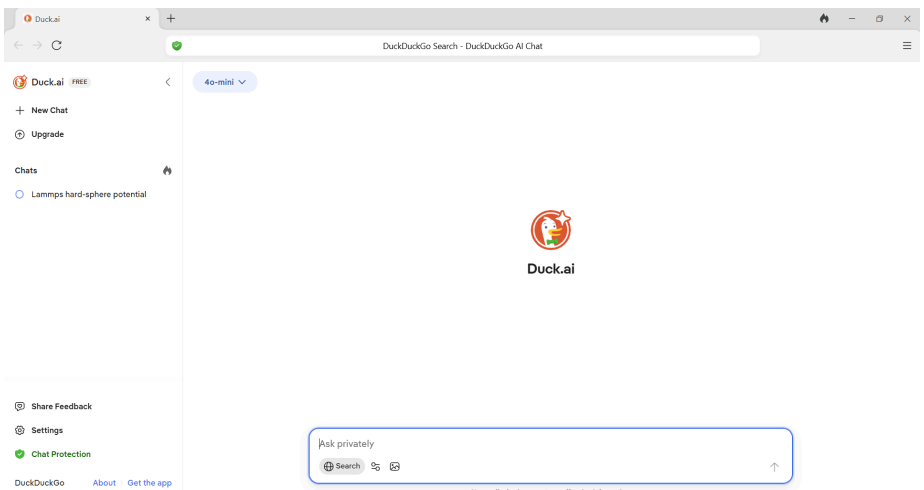
AI

“Hey Insta fam, got some next level stuff for you! 🧵 We are literally weaving the future with active fabrics that respond to environmental changes all on their own!!! ... #WeavingTheFuture #SmartTextiles (...)”

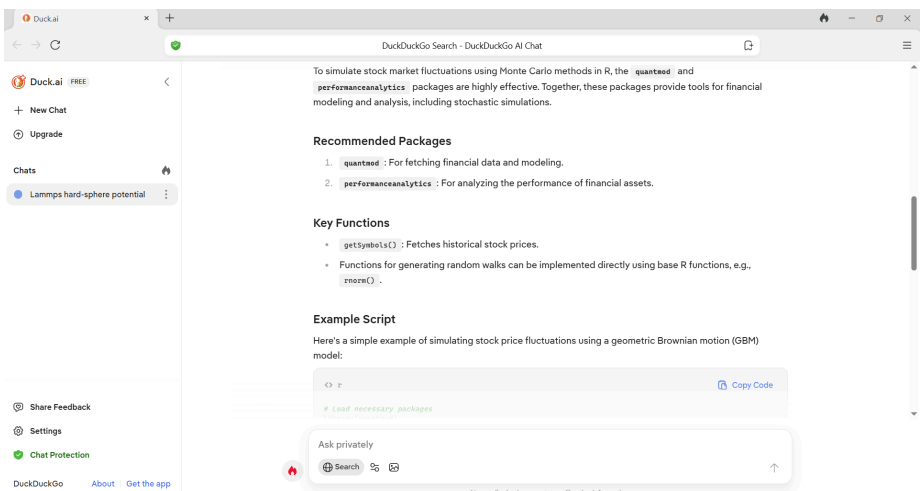
Same abstract, same model — the output changed because the **audience** was in the prompt. Write for the room you are actually talking to.

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Half the skill is the interface



prompt box · history · model · reply



the system prompt lives here

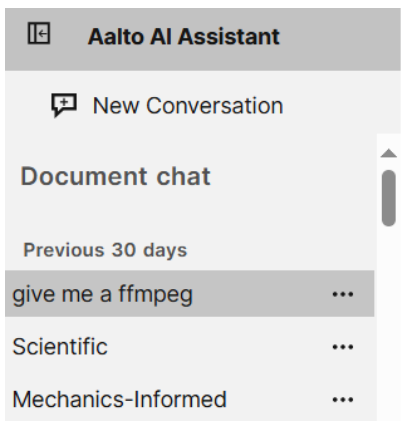
> Attach files, search the web, parameters, history: the prompt box is one of six things on this screen. Learn the screen, not the magic words.

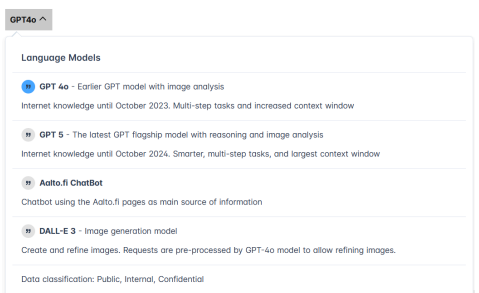
A system prompt sets the rules once

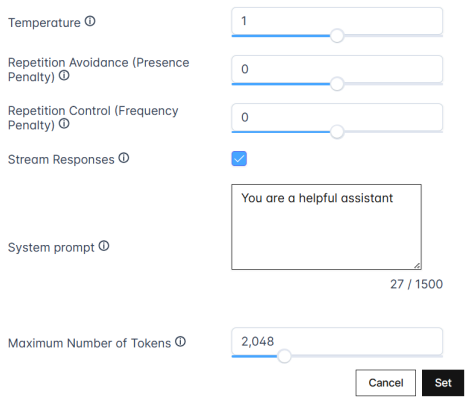
System prompt	“What is the melting temperature of water?”
You are a dino. The only words you know are graaauurr and gaurrr.	Graaauurr
You are Ronny Chieng, the Malaysian comedian … intense energy, sharp sarcasm, zero tolerance for nonsense. Don’t sugarcoat anything.	“Oh, for crying out loud, it’s 0 degrees Celsius or 32 Fahrenheit. It’s not rocket science, people.”
You are a highly capable, thoughtful, and precise assistant. Ask clarifying questions, think step by step, prioritise being truthful and nuanced, tailor to the user’s needs.	0 °C at standard pressure — plus the caveat that it varies with pressure, offered before being asked.
Be the most succinct possible in your replies.	0°C or 32°F

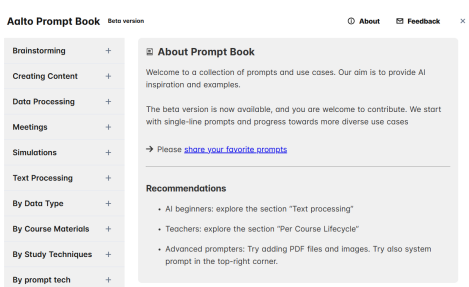
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At Aalto: the assistant that shows the data classification









history | model + data class | parameters | the Prompt Book

> Public / Internal / Confidential is printed under the model list. Read it before you paste — and suggest your own prompts in the Prompt Book.

old slides 40 + 41 · ⚠ re-shoot



Kiitos!

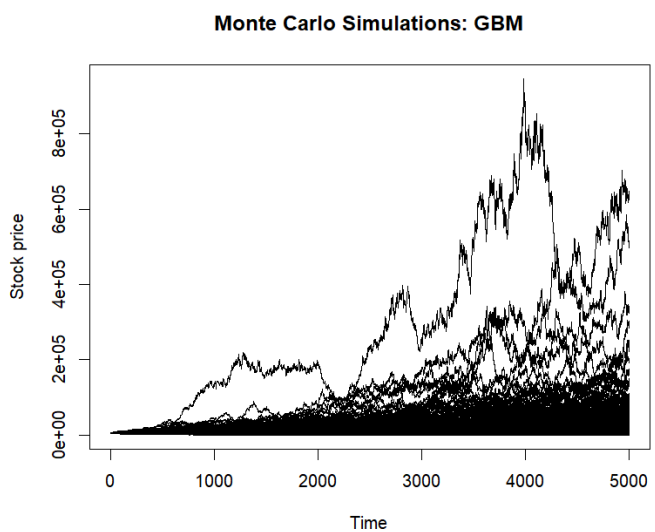
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It will also write the simulation you keep putting off

PROMPT

What R package and functions are most effective for simulating stock market fluctuations using Monte Carlo methods? Provide an example script.



the answer’s own output: 1000 GBM paths

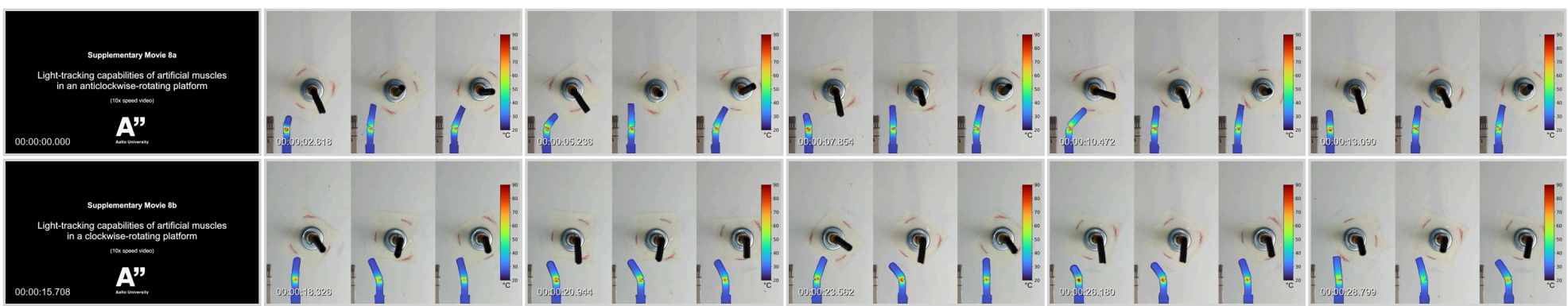
Answer: `quantmod` + `MASS`, a geometric Brownian motion loop, a seed, and a plot.

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Batch automation you can paste into a terminal

PROMPT

How can I create thumbnails for each MP4 file in a directory, using FFmpeg to select frames based on the video's duration, scale the video, apply a specific FPS value, and include a timestamp overlay?



what it produced

One command, every file in the folder — and it explains each flag after.

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